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ORIGINAL ARTICLE

Genome-wide identification and characterization of the aquaporin gene family in *Medicago truncatula*

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Abstract

Aquaporins (AQPs), or major intrinsic proteins (MIPs), constitute a large and diverse family of protein channel transporter in plants. However, the information about *AQPs* family is still poorly understood in the model legume species *Medicago truncatula*. Here, 46 *AQPs* were identified and characterized. Based on the phylogenetic analysis, these genes were assigned to five subfamilies, specifically as 10 plasma membrane intrinsic proteins, 12 tonoplast intrinsic proteins, 18 NOD26-like intrinsic proteins, 4 small basic intrinsic proteins and 2 uncharacterized intrinsic proteins. The essential information of *AQPs*, such as subcellular localization, gene structure, chromosomal localization, evolutionary stress and functional residues were systematically analyzed. In addition, in silico analysis of the gene expression profiles showed certain of tissue specificity and responsiveness to drought/salt stresses in some *AQPs* genes. The qRT-PCR results showed that *MtPIP1;5* and *MtTIP1;3* were significantly up-regulated under both drought and salinity stress conditions, indicated their important roles in response to abiotic stresses. The results could provide valuable information for the further functional analysis of the *AQP*s in *M. truncatula* and related legume species.

Keywords Aquaporins · *Medicago truncatula* · Phylogenesis · Conserved residues · Gene expression patterns · Abiotic stresses

Abbreviations

AQPs	Aquaporins
PIPs	Plasma membrane intrinsic proteins
TIPs	Tonoplast intrinsic proteins
NIPs	NOD26-like intrinsic proteins
SIPs	Small basic intrinsic proteins
XIPs	Uncharacterized intrinsic proteins
WGD	Whole-genome duplication
GRAVY	Grand average of hydropathicity
Ar/R	Aromatic/arginine
TM	Transmembrane

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Introduction

Water is essential for the growth and development of plants as it is involved in the transport of substances and a variety of physiological processes as well as the maintenance of plant morphology (Johansson et al. 2000). However, the free diffusion efficiency of water is not high enough to meet the need for rapid physiological processes, and this deficiency could be compensated by the presence of aquaporins (*AQPs*), which play an important role in water balance of plant physiology (Johansson et al. 2000). To date, numerous *AQPs* have been identified in plants compared to animals. The highest number of 120 *AQPs* has been identified in canola (Sonah et al. 2017), while, only 19 *AQP*s were identified in the primitive plant *Selaginella* *moellendorffii* (Deshmukh et al. 2015). There are seven major subfamilies of *AQPs* in plants and only five of those present in higher plants including monocot and dicots, i.e., plasma membrane intrinsic proteins (PIPs), tonoplast intrinsic proteins (TIPs), NOD26-like intrinsic proteins (NIPs), small basic intrinsic proteins (SIPs) and unclassified X intrinsic proteins (XIPs), and these subfamilies can be further divided into different subgroups (Danielson and Johanson 2008; Gupta and Sankararamakrishnan 2009; Johanson et al. 2001).

Classical *AQPs* members comprise 6 transmembrane domains (TM1–TM6) connected by five loops (LA-LE) (Maurel et al. 2008). Two half-helices (HB and HE) formed by loops B and E, which both carry a signature Asn-Pro-Ala (NPA) motif and together form a new transmembrane selective region (Maurel et al. 2008; Tornroth-Horsefield et al. 2006). In addition, the aromatic/arginine (Ar/R) selectivity filter is formed by four conserved amino acid residues, specifically as two residues from helix2 and helix5 (H2 and H5) respectively and two residues from loop E (LE1 and LE2) (Bansal and Sankararamakrishnan 2007; Maurel et al. 2008). This region is located on the extracellular end of the NPA region and it is thought to be a more important selectivity filter for the permeability of some aqua and non-aqua substrates (Bansal and Sankararamakrishnan 2007; Hove and Bhave 2011; Maurel et al. 2008; Sui et al. 2001). Beside some special binding sites, the Ar/R selectivity filter can also act as a size-exclusion barrier and may also be accompanied by the formation of hydrogen bonds and ionic interactions (Sui et al. 2001). Previous studies have reported that a spacing of a precise distance of 108 amino acids (AA) between the two NPA domains is a necessary and selective feature for silicon (Si) transporters among all Si-transporting plants (Deshmukh et al. 2015). Many other functional residues, such as the five Froger's positions (Froger et al. 1998) also play critical roles in *AQPs*.

To date, many *AQPs* have been reported to participate in various physiological and developmental processes of plants, such as seed germination, tissue expansion and formation of lateral roots (Maurel et al. 2015). The heterologous expression of *AtTIP2;1* in *Xenopus laevis* oocytes facilitated the urea transport (Liu et al. 2003). *AtNIP5;1*, *AtNIP6;1* and *OsNIP3;1* were all found to be related to the regulation of boron (Hanaoka et al. 2014; Takano et al. 2006; Tanaka et al. 2008). Some NIP subfamily members such as *AtNIP5;1*, *AtNIP6;1*, *OsNIP2;1*, *OsNIP3;2*, *LjNIP5;1* and *LjNIP6;1* were reported to facilitate the transport of arsenite (As) and antimonite (Sb) (Bienert et al. 2008). The tobacco *NtAQP1* was found to be a CO_2 channel (Uehlein et al. 2003). Meanwhile, many *AQPs* such as *NtAQP1* (Sade et al. 2010), *SlTIP2;2* (Xin et al. 2014), *TaAQP8* (Hu et al. 2012) and *TaAQP7* (Zhou et al. 2012), were found to be related to abiotic stresses such as salt and drought.

Medicago truncatula is an annual legume species with a relatively small diploid genome and a good collinearity with other legumes, and thus has long been considered as a model species for the studies of legume biology (Young et al. 2011). Furthermore, the completion of the genome sequencing of *M. truncatula* provides the possibility to comprehensively analyze this gene family in this model species. In this study, a genome-wide identification of *AQPs* gene family in *M. truncatula* was conducted, and their sequences, putative functions, evolutionary relationships, expression patterns in different tissues and responsiveness to various abiotic stresses were further characterized. This fundamental research would provide valuable information for the further physiological and functional studies of the *AQPs* family in *M. truncatula* and other related legume species.

Materials and methods

Plant growth, treatments and tissues collection

Medicago truncatula ('Jemalong A17') seeds were scarified, sterilized and germinated. The seeds were well watered and grown under greenhouse conditions (27/23 °C under a 14-h light/10 h dark cycle). The 28-day-old seedlings were treated with NaCl (180 mM), the leaves were collected 0, 6, 24 and 48 h after the treatments. Drought was imposed with 20% PEG, the leaves and roots were collected after 0, 6, 24 and 48 h of drought stress. Three independent plants were collected for each of the above time points, and immediately frozen in liquid nitrogen and stored at − 80 °C prior to use.

Identification and phylogenetic analysis of *MtAQPs* gene family in *M. truncatula*

The sequences of *M. truncatula* (gene sequences, coding sequences (CDS) and protein sequences) were retrieved from the *Medicago truncatula* Genome Database (http://www.medicagogenome.org/). The *AQPs* sequences (nucleotide and amino acid) of *A. thaliana*, *P. vulgaris* and *P. trichocarpa* were retrieved from Phytozome database (https://phytozome.jgi.doe.gov/pz/portal.html). The 35 *AQPs* sequences of *A. thaliana* and 55 of *P. trichocarpa* were used to extract the *AQP*-like genes of *M. truncatula* using the Blastp and Tblastn programs respectively by BLAST 2.4.0 tool. For the gene with different isoforms, only the primary one was retained by referring to the Phytozome database. The functional domains were predicted by the Conserved Domain Database (Marchlerbauer et al. 2015) (CDD) of NCBI and SMART analysis (Letunic et al. 2015). All the obtained protein sequences were examined for the presence of AQP domains using the Pfam (http://pfam.sanger.ac.uk/search) tool. Meanwhile, in order to confirm the accuracy of the putative genes, the PF00230 (major intrinsic protein) was used to search the *M. truncatula* genome at Phytozome database. Multiple sequence alignments of *AQPs* of four species, *M. truncatula*, *A. thaliana*, *P. vulgaris* and *P. trichocarpa*, were performed using the ClustalW program in MEGA 7 software (Kumar et al. 2016). The unrooted phylogenetic tree was constructed by MEGA 7 based on the maximum likelihood method and the stability of the branch node was measured by performing 1000 bootstraps. The nomenclature of *MtAQPs* was based on the homology with the clear aquaporins of three other species following the rules proposed previously (Johanson et al. 2001).

Sequence analysis and in silico prediction of subcellular localization of *MtAQPs*

Gene structure was displayed using the Gene Structure Display Server (GSDS 2.0) (http://gsds.cbi.pku.edu.cn/) (Hu et al. 2015). The transmembrane helix and topology of the MtAQPs were predicted by MEMSAT-SVM prediction method in Phyre2 server (http://www.sbg.bio.ic.ac.uk/phyre2/html/page.cgi?id=index). Molecular weight, theoretical isoelectric point and grand average of hydropathicity (GRAVY) of each *AQPs* was calculated by the ProtParam tool of ExPASy Sever (http://web.expasy.org/protparam/). For estimation of divergence time, the Synonymous (Ks) and nonsynonymous substitution (Ka) substitution rates were calculated. Protein sequences of the gene pairs were multiply aligned using Clustal X program. Then, the aligned amino acid sequences and their corresponding cDNA sequences were analyzed using the CODEML program in PAML interface tool of PAL2NAL programs using the pairwise sequences with more than 80% similarity (Suyama et al. 2006). The subcellular localization of *AQPs* was performed using the Plant-mPLoc (Kuochen and Shen 2010) and WoLF PSORT (http://www.genscript.com/wolf-psort.html) algorithms.

Chromosomal locations and gene duplication analysis

The chromosomal localization was displayed using the MapChart 2.3 software. Collinear blocks were evaluated by MCScan (http://chibba.pgml.uga.edu/mcscan2/), and the alignments with E-value $\leq$ 1E$-$10 were considered as significant matches (Tang et al. 2008). Tandem duplicated genes were defined as adjacent homologous genes on a single chromosome, with no more than one intervening gene (Zhu et al. 2014).

In silico analysis of promoter sequences, conserved amino acid residues and gene expression patterns

The 2000 bp upstream nucleotide sequences from the initiation code of genes were downloaded from the *M. truncatula* Genome Database, and analyzed by the PlantCARE (Lescot et al. 2002). The functional analysis of conserved amino acid residues including the dual NPA motifs, Ar/R selectivity filter (H2, H5, LE1 and LE2), and five Froger's positions (P1–P5) (Guenther et al. 2003; Hove and Bhave 2011; Tornroth-Horsefield et al. 2006; Zou et al. 2015a). The multiple sequence alignments in this work were performed using the Clustal Omega program (http://www.ebi.ac.uk/Tools/msa/clustalo/). The *AQPs* gene chip expression data was retrieved from the *M. truncatula* Gene Expression Atlas (MtGEA) Project of NOBLE database (https://mtgea.noble.org/v3/) (Benedito et al. 2008).

RNA isolation and real-time reverse transcription PCR (qRT-PCR) analysis

Total RNA was isolated from 25 to 50 mg of frozen tissue using the Sangon UNIQ-10 column TRIzol total RNA extraction kit and treated with DNase (TaKaRa, Dalian, China) to remove DNA contamination. Subsequently, first-strand cDNA was synthesized from 4 μg of total RNA using the M-MuLV reverse transcriptase according to the manufacturer's instructions (Sangon Biological Engineering Technology & Services, China). The specific primers (Table S1) for the qRT-PCR analysis were designed using the Primer3 software, and their specificity was checked using information provided on the NCBI website. The qRT-PCR was performed in three technical replicates for each biological duplicate using the 7500 Fast Real-Time PCR system (Applied Biosystems, USA). Each 20 μL reaction volume included 10 μL of 2 × SG Fast qPCR Master Mix (Low Rox), 2 μL of DNF Buffer, 0.5 μL of the forward and reverse primers, 1 μL of diluted cDNA solution, and 6 μL of ddH_2O. The thermal profile used for all PCR amplifications was as follows: 10 min at 95 °C for DNA polymerase activation, followed by 40 cycles of 15 s at 95 °C and 1 min at 60 °C. As an internal standard, *Mt-Actin* was used to calculate the relative fold differences based on the comparative *Ct* method. The expression levels were calculated using the $2^{-\Delta\Delta CT}$ method.

Results and discussion

Identification, classification and phylogenetic analysis of *MtAQPs*

In this study, a total of 46 putative loci encoding complete *AQP*-like genes were found, and further named as *MtAQPs* following the rules proposed previously (Johanson et al. 2001) (Table 1). The number of *MtAQPs* are greater than that in some species such as *A. thaliana*, *Physcomitrella patens*, maize and rice (Table S2). Previous studies have showed that more *AQPs* contained in species such as poplar, soybean, chickpea, rubber tree, Chinese cabbage and upland cotton were attributed to the whole-genome duplication (WGD) events (Table S2). Similarly, a WGD event played a significant role to *M. truncatula* genome approximately 58 million years ago, and after the WGD, the *M. truncatula* genome experienced a high level of rearrangement (Young et al. 2011).

In this study, by phylogenetic analysis with three other identified species (Table S2), the *AQPs* genes of *M. truncatula* were classified into five subfamilies, i.e., 10 PIPs, 12 TIPs, 18 NIPs, 4 SIPs and 2 XIPs (Table 1; Fig. 1), and the top four subfamilies were further divided into 2, 5, 7 and 2 subgroups, respectively, the XIP subfamily had only one subgroup with 2 members. In contrast to other species, *M. truncatula* has fewer subgroups in XIP subfamily, maybe the gene loss occurred in this subfamily. The same case has also been found in some species such as common bean (Ariani and Gepts 2015), soybean (Zhang et al. 2013) and physic nut (Zou et al. 2016). Previously study showed that the XIPs were completely missing or have not yet been reported in species from a galegoid clade of papilionoideae family such as Medicago and *Lotus japonicus* (Deokar and Tar'an 2016; Deshmukh and Bélanger 2016). But in this study, two XIP genes were identified in *M. truncatula*, this observation suggests that the XIPs genes were not only existed in warm season legumes but also in the cool-season legumes during the evolution of the papilionoideae family. Furthermore, the splice variants of all *MtAQPs* were predicted, and only one gene (*MtPIP1;1*) might alternatively spliced to form four putative proteins in PIP subfamily. The clear evolutionary relationships of the *AQPs* family can be further observed from the evolutionary divergence between sequences (Table S4).

The genetic Ka/Ks ratios of 13 gene pairs were all less than 0.5 (Table S5), the lowest Ka/Ks ratio is only 0.0468 (*MtPIP1;1/MtPIP1;2*), and the highest Ka/Ks ratio is 0.4902 (*MtNIP1;3/MtNIP1;4*). These findings suggest that the *AQPs* has undergone strong purifying selection and are slowly evolving at the protein level. The putative protein length of these genes varied from 218 AA (*MtNIP6;3*) to 329 AA (*MtXIP1;1*) with an average of 270 AA residues. Molecular weights varied from 22,504.37 Da (*MtNIP6;3*) to 35,355.68 Da (*MtXIP1;1*), and the isoelectric points varied from 4.8 (*MtTIP2;3*) to 9.83 (*MtSIP1;3*). A great divergence of the GRAVY was observed, varied from 0.026 (*MtNIP7;1*) to 0.947 (*MtTIP2;3*), with the average GRAVY of 0.43, 0.81, 0.46, 0.69 and 0.71 in five subfamilies, respectively. In *AQP*s, identification of the TMs is difficult with the available tools because of the two half helixs harboring the NPA motifs, so the homology-based 3D modeling was used to predicted TMs by MEMSAT-SVM prediction method in Phyre2 in our study. Nearly 95.7% proteins possessed six helixes except for *MtNIP6;4* and *MtSIP1;3*, which only contain three and seven helixes respectively (Table 2). The similar case also reported in common bean (Ariani and Gepts 2015), soybean (Zhang et al. 2013) and tomato (Reuscher et al. 2013).

The analysis of functional domains indicated that all putative genes had the motifs and domains that specific for the *AQPs* family. However, two genes showed some special properties. Concretely, there is a 63 aa heavy-metal-associated (HMA) domain (PF00403) at the intermediate region of the *MtNIP6;2* sequence, which contains two conserved cysteines that are important in binding and transfer of metal ions (Bull and Cox 1994). Besides, this gene harbors the very low GRAVY (0.049) and the lowest number of transmembrane helixes (4), meanwhile, high divergence of functions related residues can be observed (Table 2). In brief, this gene is extremely hydrophilic and may act as a metal associated transporter, whereas the role of a normal *AQPs* may be weakened or has some special functions. The other one, *MtNIP7;1*, which is so long (813 aa) that has the space to accommodate another domain, i.e., the zeta_toxin domain (PF06414), a member of P-loop NTPase superfamily, is thought to be part of a postregulational killing system in bacteria and related to programmed cell death in prokaryotes (Meinhart et al. 2003). Besides, it has the most seven helixes and the lowest GRAVY (0.026). These features revealed that this protein might exhibit special and dual functions.

Genomic organization and in silico analysis of subcellular localization of MtAQPs

Forty-six *MtAQPs* were distributed unevenly on eight chromosomes, the chromosome 4 has the most genes (10) and the chromosome 6 has the least (2). In addition, based on the clusters, seven genes were observed on chromosome 1 (2 genes), 2 (3 genes) and 4 (2 genes) (Fig. 2). The genes clustered together at a certain region on a chromosome were highly similar and belong to the same subgroup, which indicated that these genes may have similar transport functions. Meanwhile, some genes with high similarity also

Table 1 The details of *AQPs* gene family identified in *M. truncatula*

S. no.	Gene name	Phytozome ID	CDS/bp	Protein/AA	Chromosomal location	Molecular weight/D	*p*I	GRAVY	Comments
1	MtPIP1;1	Medtr3g070210.1	873	290	chr3:31461214..3145889	31,066	8.68	0.357	
2	MtPIP1;2	Medtr5g082070.1	870	289	chr5:35231538..3522921	30,926.09	8.61	0.440	
3	MtPIP1;3	Medtr8g098375.1	864	287	chr8:41036621..4103848	30,958.94	9.00	0.367	
4	MtPIP1;4	Medtr5g010150.1	864	287	chr5:2650689..2648664	30,799.71	8.97	0.350	
5	MtPIP1;5	Medtr4g415300.1	870	289	chr4:4483857..4480747	31,159.86	8.60	0.253	
6	MtPIP2;1	Medtr1g095070.1	666	221	chr1:42795577..4279928	23,550.92	9.48	0.840	
7	MtPIP2;2	Medtr2g094270.1	864	287	chr2:40208588..4020606	30,806.54	7.64	0.382	
8	MtPIP2;3	Medtr4g059390.1	858	285	chr4:21792505..2179497	30,512.11	6.58	0.368	
9	MtPIP2;4	Medtr7g101190.1	864	287	chr7:40836017..4083426	30,657.52	8.60	0.425	
10	MtPIP2;5	Medtr3g118010.1	840	279	chr3:55220327..5522156	29,759.87	9.26	0.525	
11	MtTIP1;1	Medtr7g103030.1	753	250	chr7:41419534..4141735	25,393.41	5.60	0.854	
12	MtTIP1;2	Medtr1g115405.1	759	252	chr1:52103643..5210213	26,129.19	6.01	0.797	
13	MtTIP1;3	Medtr7g023350.1	759	252	chr7:7654628..7656988	26,017.02	5.30	0.763	
14	MtTIP1;4	Medtr4g063090.1	759	252	chr4:23338918..2334103	25,800.88	5.41	0.829	
15	MtTIP2;1	Medtr2g101370.1	750	249	chr2:43584848..4358584	25,241.3	6.15	0.904	
16	MtTIP2;2	Medtr8g013680.1	747	248	chr8:4191032..4187967	25,158.16	5.29	0.906	
17	MtTIP2;3	Medtr5g012810.1	744	247	chr5:3928871..3930292	24,927.88	4.80	0.947	
18	MtTIP3;1	Medtr6g069040.1	768	255	chr6:24832348..2483438	27,106.65	6.79	0.684	
19	MtTIP3;2	Medtr1g077540.1	777	258	chr1:34607915..3460637	27,568.18	7.86	0.594	
20	MtTIP4;1	Medtr1g006490.1	744	247	chr1:44430..46280	25,769.98	5.64	0.810	
21	MtTIP4;2	Medtr1g006590.1	744	247	chr1:46548..50437	25,853.37	6.26	0.864	
22	MtTIP5;1	Medtr2g060360.1	765	254	chr2:24861455..2485952	26,550.81	7.75	0.776	
	PseudoTIP5;2	Medtr6g044390.2	258	85	chr6:15339500..1534120	8764.02	4.46	0.656	Both NPAs missing
	PseudoTIP5;3	Medtr8g468730.1	303	100	chr8:24918865..2491922	10,371.97	4.46	0.757	Both NPAs missing
23	MtNIP1;1	Medtr2g017570.1	822	273	chr2:5388011..5391282	29,045.42	7.12	0.392	
24	MtNIP1;2	Medtr2g017610.1	822	273	chr2:5401931..5404382	29,078.59	7.09	0.417	
25	MtNIP1;3	Medtr2g017590.1	813	270	chr2:5396335..5398792	28,619.4	6.58	0.592	
26	MtNIP1;4	Medtr2g017620.1	813	270	chr2:5409004..5406235	28,764.42	7.83	0.507	
27	MtNIP1;5	Medtr8g087710.1	831	276	chr8:36261986..3626440	29,408.04	7.06	0.485	
28	MtNIP1;6	Medtr8g087720.1	810	269	chr8:36267093..3626969	28,572.15	9.16	0.507	
29	MtNIP1;7	Medtr2g017660.1	831	276	chr2:5414825..5416885	29,988.8	5.87	0.463	
30	MtNIP2;1	Medtr7g010650.1	825	274	chr7:2666062..2661639	28,891.37	8.73	0.350	
31	MtNIP3;1	Medtr5g063930.1	789	262	chr5:26519938..2651798	27,796.53	6.64	0.627	
32	MtNIP4;1	Medtr4g084880.1	822	273	chr4:33127385..3312897	29,280.28	7.67	0.629	
33	MtNIP4;2	Medtr3g008920.1	795	264	chr3:1704466..1703370	28,117.06	8.20	0.683	
34	MtNIP4;3	Medtr5g079570.1	807	268	chr5:34018177..3402006	28,553.16	5.47	0.604	
35	MtNIP5;1	Medtr1g097840.1	903	300	chr1:44096826..4410217	31,031.97	8.77	0.500	
36	MtNIP6;1	Medtr2g104940.1	933	310	chr2:45234196..4523171	32,251.56	8.77	0.359	
37	MtNIP6;2	Medtr4g006650.1	969	322	chr4:736209..733098	35,198.28	6.77	0.049	
38	MtNIP6;3	Medtr4g006630.1	657	218	chr4:716770..720058	22,504.37	8.60	0.730	
39	MtNIP6;4	Medtr4g006730.1	918	305	chr4:777723..773587	31,683.91	8.32	0.444	
40	MtNIP7;1	Medtr1g070385.1	2442	813	chr1:31163507..3117149	90,389.87	8.98	0.026	
41	MtSIP1;1	Medtr7g085070.1	741	246	chr7:32887703..3289275	26,584.5	9.13	0.696	
42	MtSIP1;2	Medtr6g033010.1	753	250	chr6:10341902..1033747	27,301.34	9.28	0.655	
43	MtSIP1;3	Medtr4g049340.1	720	239	chr4:17328473..1732725	26,010	9.83	0.682	
44	MtSIP2;1	Medtr7g088310.1	720	239	chr7:34383893..3438783	26,387.37	9.45	0.723	

Table 1 (continued)

S. no.	Gene name	Phytozome ID	CDS/ bp	Protein/ AA	Chromosomal location	Molecular weight/D	*p*I	GRAVY	Comments
45	MtXIP1;1	Medtr4g075570.1	990	329	chr4:28881013..2888229	35,355.68	6.29	0.665	
46	MtXIP1;2	Medtr4g075590.1	951	316	chr4:28885867..2888730	33,507.54	6.04	0.764	

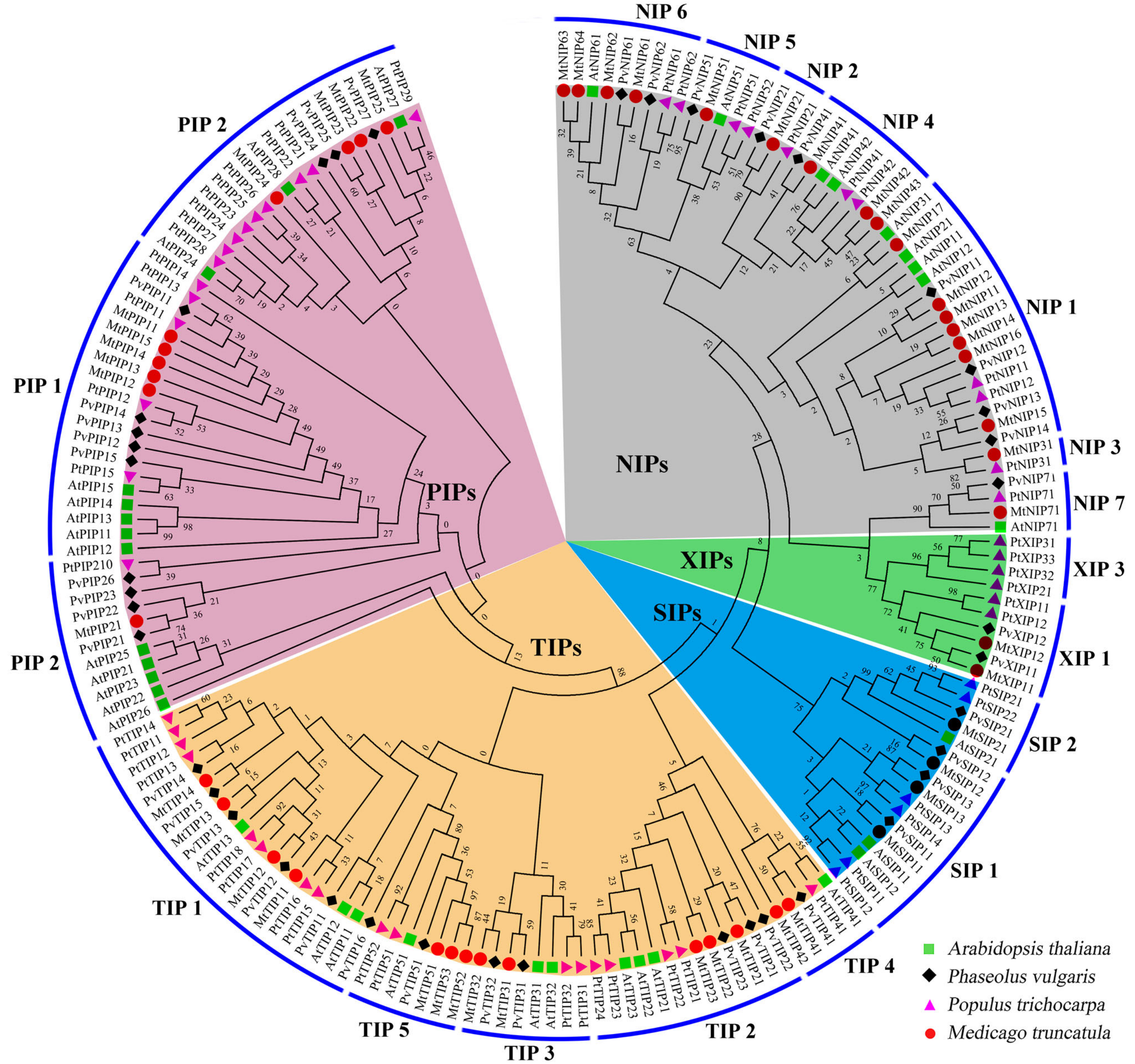

Fig. 1 Phylogenetic relationship among the AQPs proteins of *A. thaliana*, *P. vulgaris*, *P. trichocarpa* and *M. truncatula*. The AQPs protein sequences of four species were aligned using the ClustalW program implemented in MEGA 7 software. The red dot, green rectangles, black rhombuses, and purple triangle color, represent *M. truncatula*, *A. thaliana*, *P. vulgaris*, and *P. trichocarpa*, respectively. And the nomenclature of *MtAQPs* were based on homology with the identified aquaporins that clustered together

Table 2 List of *MtAQPs* along with their corresponding functional residues, transmembrane helixes (TM), subcellular localizations and the number of introns and exons

Gene name	TM[a]	Introns	Exons	Plant-mPLoc[b]	WoLF PSORT[b]	NPA motifs		Ar/R selectivity filter				Froger's positions				
						LB	LE	H2	H5	LE1	LE2	P1	P2	P3	P4	P5
MtPIP1;1	6	3	4	Plas	Plasb	NPA	NPA	F	H	T	R	E	S	A	F	W
MtPIP1;2	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	E	S	A	F	W
MtPIP1;3	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	E	S	A	F	W
MtPIP1;4	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	E	S	A	F	W
MtPIP1;5	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	Q	S	A	F	W
MtPIP2;1	6	3	4	Plas	Vacu	NPA	NPA	L	H	T	R	Q	S	A	F	W
MtPIP2;2	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	Q	S	A	Y	W
MtPIP2;3	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	Q	S	A	F	W
MtPIP2;4	6	3	4	Plas	Plas	NPA	NPA	F	H	T	R	Q	S	A	F	W
MtPIP2;5	6	1	2	Plas	Plas	NPA	NPA	F	H	T	R	M	S	A	F	W
MtTIP1;1	6	1	2	Vacu	Cyto	NPA	NPA	H	I	A	V	T	S	A	Y	W
MtTIP1;2	6	2	3	Vacu	Vacu	NPA	NPA	H	I	A	V	T	S	A	Y	W
MtTIP1;3	6	2	3	Vacu	Vacu/cyto	NPA	NPA	H	I	A	V	T	S	A	Y	W
MtTIP1;4	6	2	3	Vacu	Vacu	NPA	NPA	H	I	A	V	T	S	A	Y	W
MtTIP2;1	6	2	3	Vacu	Cyto	NPA	NPA	H	I	G	R	T	S	A	Y	W
MtTIP2;2	6	2	3	Vacu	Vacu	NPA	NPA	H	I	G	R	T	S	A	Y	W
MtTIP2;3	6	2	3	Vacu	Vacu	NPA	NPA	H	I	G	R	T	S	A	Y	W
MtTIP3;1	6	2	3	Vacu	Mito	NPA	NPA	H	I	A	L	T	A	S	F	W
MtTIP3;2	6	2	3	Vacu	Chlo	NPA	NPA	H	I	A	R	T	A	A	F	W
MtTIP4;1	6	2	3	Vacu	Vacu	NPA	NPA	H	I	A	R	S	S	A	Y	W
MtTIP4;2	6	2	3	Vacu	Vacu	NPA	NPA	N	I	A	R	T	S	A	F	W
MtTIP5;1	6	2	3	Plas/vacu	Chlo	NPA	NPA	S	V	G	C	V	A	A	Y	W
PseudoTIP5;2	6	1	2	Plas	Vacu	–	–	–	V	–	–	–	–	–	–	–
PseudoTIP5;3	3	1	2	Plas	Vacu	–	–	–	V	G	–	–	–	–	Y	W
MtNIP1;1	6	4	5	Plas	Plas	NPA	NPA	W	V	A	R	F	S	A	Y	L
MtNIP1;2	6	4	5	Plas	Plas	NPA	NPA	W	V	A	R	F	S	A	Y	L
MtNIP1;3	6	4	5	Plas	Plas	NPA	NPA	W	V	A	R	F	S	A	Y	M
MtNIP1;4	6	4	5	Plas	Vacu	NPA	NPA	W	V	A	R	F	S	A	Y	M
MtNIP1;5	6	4	5	Plas	Plas	NPA	*NPV*	W	V	A	R	F	T	A	Y	L
MtNIP1;6	6	4	5	Plas	Vacu	NPA	NPA	W	V	A	R	F	S	A	Y	L
MtNIP1;7	6	4	5	Plas	Plas	NPA	NPA	W	I	A	R	F	S	A	Y	L
MtNIP2;1	6	4	5	Plas	Plas	NPA	NPA	G	S	G	R	L	T	A	Y	M
MtNIP3;1	6	4	5	Plas	Plas	*NPS*	NPA	S	I	A	R	Y	S	A	Y	I
MtNIP4;1	6	4	5	Plas	Plas	NPA	NPA	W	V	A	R	F	T	A	Y	I
MtNIP4;2	6	0	1	Plas	Plas	NPA	NPA	W	L	A	R	F	S	A	Y	I
MtNIP4;3	6	4	5	Plas	Plas	NPA	NPA	W	V	G	R	F	S	A	Y	V
MtNIP5;1	6	3	4	Plas	Plas	*NPS*	*NPV*	A	I	G	R	Y	T	A	Y	L
MtNIP6;1	6	4	5	Plas	Plas	NPA	*NPV*	T	I	A	R	F	T	A	Y	L
MtNIP6;2	6	5	6	Plas	Plas/cyto	NPA	*EPL*	T	V	E	K	F	V	V	E	L
MtNIP6;3	6	3	4	Plas	Plas/vacu	NPA	*NPV*	T	I	A	R	F	T	A	Y	L
MtNIP6;4	3	4	5	Plas	Plas	NPA	*NPV*	T	I	A	R	F	T	A	Y	L
MtNIP7;1	6	11	12	Plas/chlo	Plas	NPA	NPA	A	V	G	R	Y	S	A	Y	I
MtSIP1;1	6	2	3	Plas/vacu	Vacu	*NPT*	NPA	V	V	P	N	I	A	A	Y	W
MtSIP1;2	6	2	3	Plas/vacu	Vacu	*NPT*	*NPI*	I	I	P	L	I	A	A	Y	W
MtSIP1;3	7	0	1	Plas/vacu	Chlo	*NPS*	NPA	N	V	P	N	I	A	A	F	W
MtSIP2;1	6	2	3	Plas	Vacu	*NPL*	NPA	S	H	G	A	I	V	A	Y	W

Table 2 (continued)

Gene name	TM[a]	Introns	Exons	Plant-mPLoc[b]	WoLF PSORT[b]	NPA motifs		Ar/R selectivity filter				Froger's positions				
						LB	LE	H2	H5	LE1	LE2	P1	P2	P3	P4	P5
MtXIP1;1	6	1	2	Plas	Plas	*SPV*	NPA	V	V	V	R	M	C	A	F	W
MtXIP1;2	6	1	2	Plas	Plas	*SPV*	NPA	V	V	V	R	M	C	A	F	W

The motifs that different from the general "NPA" motif were marked italic

Plas, plasma membrane; Vacu, vacuolar membrane; Chlo, chloroplast; Cyto, cytosol; Mito, mitochondria

[a]Transmembrane helixes predicted by MEMSAT-SVM prediction method in Phyre2 server

[b]The most probable subcellular localizations of AQPs based on two programs

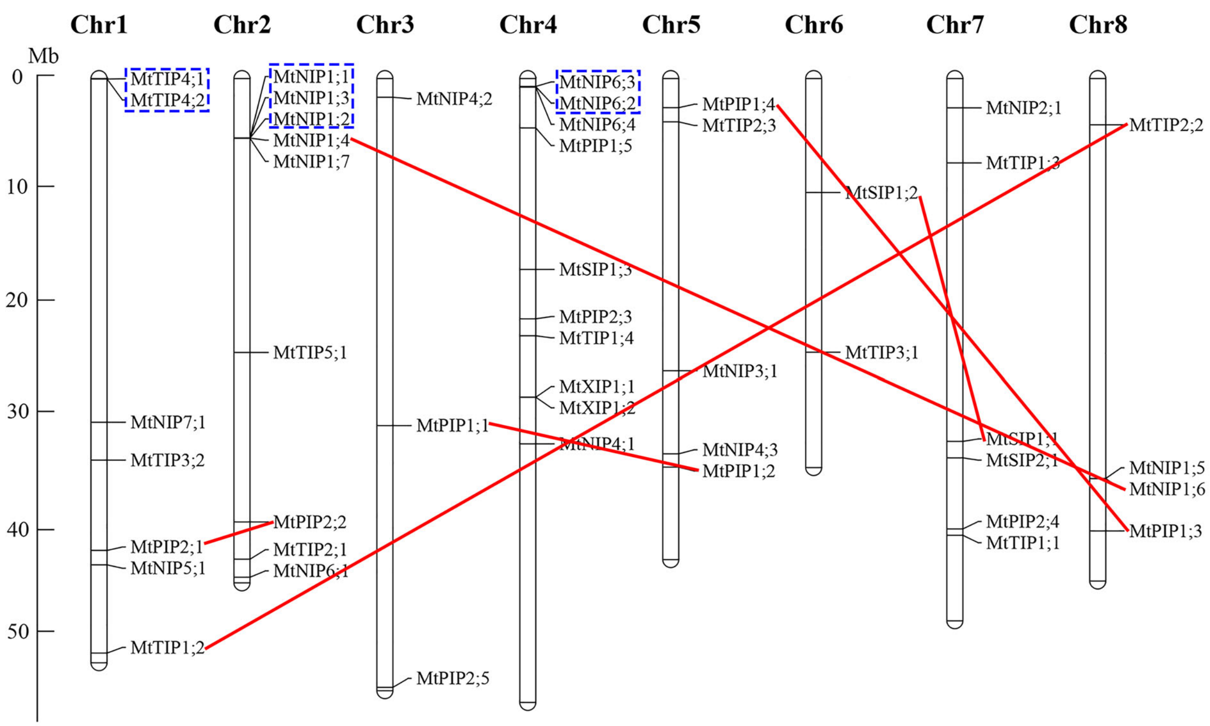

Fig. 2 Chromosomal localization of 46 *MtAQP*s on 8 chromosomes. The position of each gene shown on the graph is the average value of the two ends of the gene, and size of the chromosome is represented using a vertical scale (Mb). Three clusters of MtAQP genes are indicated in blue boxes, and red lines indicate duplicated gene pairs (colour figure online)

found located on different chromosomes, which may be related to the gene duplication events, and this may be a strategy for the plant to ensure the stable inheritance of the protein family due to its vital role. During the evolution of a gene family, new genes with novel functions may occur as a result of various processes, such as tandem duplication and WGD or segmental duplication (Roth et al. 2007). To clarify the potential evolutionary mechanism of the *MtAQPs* gene family, the MCscanX software was used to investigated the duplication events of *MtAQPs* gene family in the whole *M. truncatula* genome. In total, 7 (15.2%) *AQP* genes were assigned to tandem duplication and 10 (21.7%) *AQP* genes were found to be segmentally duplicated (Fig. 2). These results indicated that some *AQP* genes increased rapidly during evolution, and the segmental duplication might play a key role in the expansion of this family together with tandem duplications.

The subcellular localization analysis showed that most *MtAQPs* were localized to plasma membrane or vacuolar membrane (Table 2), this situation was also observed in many other species such as common bean (Ariani and Gepts 2015), soybean (Zhang et al. 2013), rubber tree (Zou et al. 2015a), and chickpea (Deokar and Tar'an 2016) and may be an ancient feature that related to specific functions of *AQPs*. The members of PIP, NIP and XIP subfamilies were mainly localized to the plasma membrane, whereas the members of TIP subfamily mainly to vacuolar membrane. Changeably, the members of SIP subfamily were predicted to localize to the plasma membrane or vacuolar membrane. Obviously, the PIP and TIP subfamilies have their special feature that they tend to be localized to where the subfamily name describes. Localizing to other putative positions such as chloroplast, mitochondria and cytosol were also observed, which indicated that there may be a diverse and complex positioning signal system in *AQPs* sequences and may be related to corresponding important transport functions.

Structural features and promoter sequences analysis of *MtAQPs*

The number of exons mainly ranged from two to four (93%), the highest number of 12 exons were identified in *MtNIP7;1*, while, only one exon was observed in *MtNIP4;2* and *MtSIP1;3*. Although the length of *MtAQPs* varied in a wide range, the length of each CDS and protein is similar within each subfamily. In addition, the number of exons and introns were also conserved within each subfamily (Table 2; Fig. 3). The PIP subfamily was characterized by four exons (90%), except for *MtPIP2;5*, which has two exons. The TIP subfamily was characterized by three exons (91%), except for *MtTIP1;1*, which also has two exons. The NIP subfamily features five exons (72%) except for five genes (*MtNIP4;2*, *MtNIP5;1*, *MtNIP6;2*, *MtNIP6;3* and *MtNIP7;1*), of which *MtNIP4;2* has no intron and *MtNIP7;1* has the most exons of 12. The SIP subfamily features 3 exons (75%), which was similar with the TIP subfamily, and *MtSIP1;3* of this subfamily has no intron. The two members of XIP subfamily both have the least exons of two (100%). The number of exons within each subfamily was coincident with that of *AQPs* of many other species (Ariani and Gepts 2015; Deokar and Tar'an 2016; Zhang et al. 2013; Zou et al. 2015a). However, the number of exons in sequences of XIP subfamily members was mainly observed by two or three in different species, i.e., *M. truncatula* and soybean XIP members all have two exons (Zhang et al. 2013), whereas tomato XIP subfamily features three exons (Reuscher et al. 2013). The conservation of the *AQPs* gene structure revealed the ancient features of the evolution.

Locations of *cis*-acting regulatory elements in the promoter sequences of *MtAQPs* were predicted to understand the possible roles of *MtAQPs* in response to abiotic stresses (Table S6). Finally, six elements were selected for further analysis, i.e., abscisic acid responsive elements (ABRE), elements essential for the anaerobic induction (ARE), low-temperature responsive elements (LTR), drought induced elements (MBS), defense and stress responsive elements (TC-rich repeats) and circadian control elements (Circadian). Based these data, nearly 91%, 74%, 50%, 37%, 30% and 15% *MtAQPs* contain at least one ARE, ABRE, MBS, LTR, TC-rich repeats and Circadian elements, respectively. Interestingly, light responsive elements were observed to present in all *MtAQPs* with diverse kinds (data not shown), such as ACE, Box I, G-Box, GT1-motif and I-Box etc. The same case was also reported in *AQPs* of chickpea (Deokar and Tar'an 2016) and which may be a common feature shared by different species. Meanwhile, the presence of ARE, ABRE, MBS and LTR elements in many *MtAQPs* indicated that these genes may be involved in some stress response. This strategy may be well suited to the plant growth habits, namely, it may be important to the photosynthesis and some other physiological processes of the plant. The transcripts of two *Arabidopsis* genes *AtPIP1;2* and *AtPIP2;1* were affected by circadian rhythm (Takase et al. 2011). In addition, some *AQPs* were reported to respond to drought or low temperature and contain correlative *cis*-acting elements MBS or LTR, such as *CaTIP1;4* and *CaNIP4;1* of chickpea (Deokar and Tar'an 2016) and *EgTIP2* of *Eucalyptus grandis* (Rodrigues et al. 2013). These abiotic stresses related *cis*-acting elements in promoter sequences might confer corresponding resistance to these *MtAQPs*. However, the opposite case was also observed, the application of exogenous ABA induced a significant decrease in the transcripts of *EgTIP2* of *E. grandis* (Rodrigues et al. 2013), which has at least two ABA related *cis*-elements (ABRE core and MOTIF IIb).

Functions related conserved residues analysis of *MtAQPs*

Aquaporins were known for transporting water and other diverse solutes, whereas the structures of *MtAQPs* were strongly characterized by the conservation of some residues within each subfamily even in the whole *MtAQPs* family (Table 2 and Supplementary S1). The general domain, two NPA motifs, were conserved in all *MtAQPs* of PIP and TIP subfamilies. Sometimes the substitutions of one or two amino acid residues of this motif may also occur, such as some members of NIP, SIP and XIP subfamilies. The same case occurred in *AtNIP5;1* and *AtNIP6;1* but they were functional (Hove and Bhave 2011). Meanwhile, the adjacent residues of the two NPA motifs were relatively conserved within each subfamily, which were all considered to be important to the formation of the NPA filter. Obviously, its selectivity of substrates was limited for it is ubiquitous (Hove and Bhave 2011). The comparative analysis of the Ar/R (aromatic/arginine) filters of *AQPs* of four legumes (Table S7), i.e., common bean (Ariani and Gepts 2015), soybean (Zhang et al. 2013),

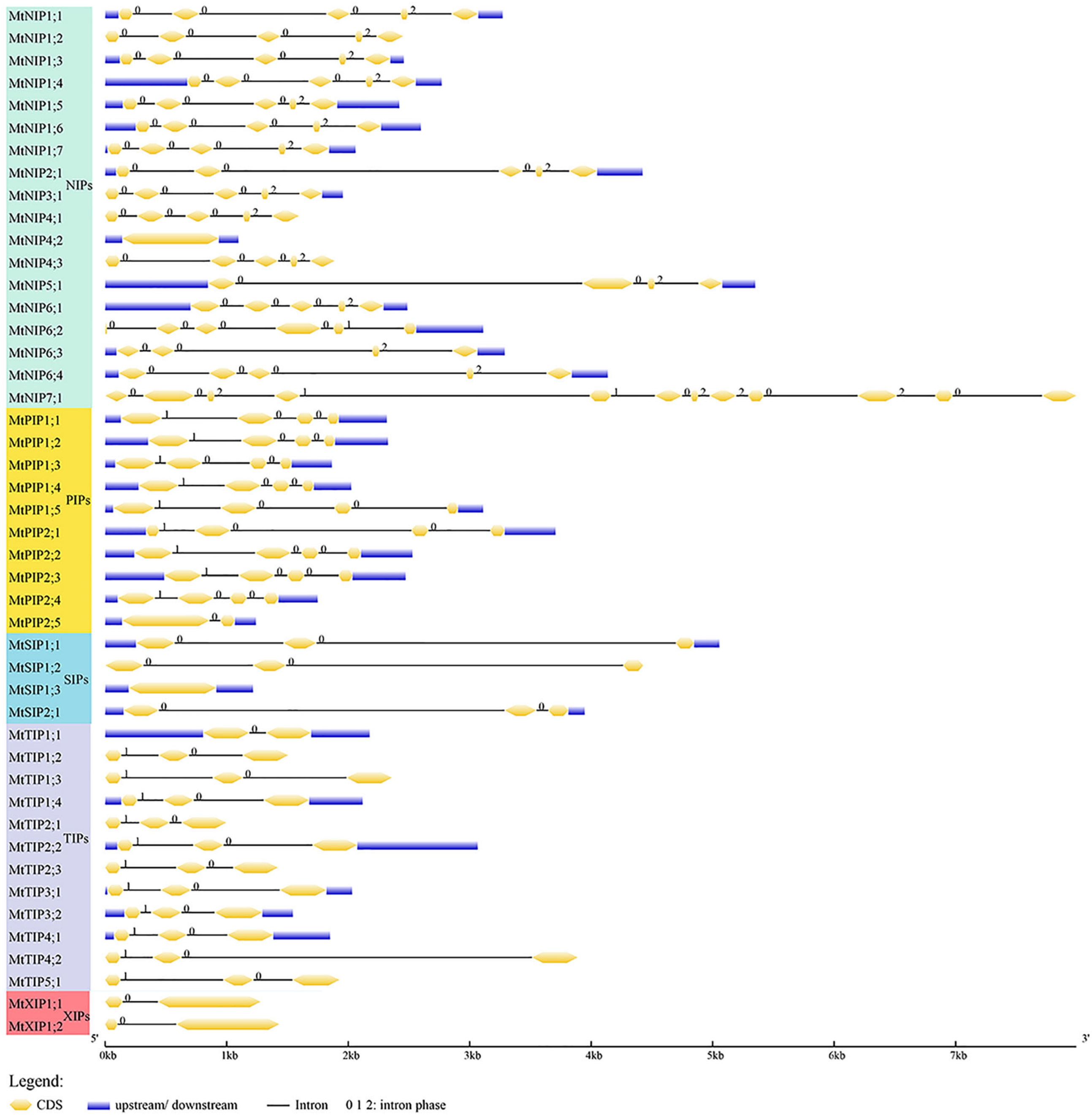

Fig. 3 Gene exon–intron structures of *AQPs* family members in *M. truncatula*. Exons and introns are represented by yellow boxes and black lines, respectively. The numbers 0, 1, and 2 represent intron phases (colour figure online)

chickpea (Deokar and Tar'an 2016) and *M. truncatula*, indicated that all PIP members were characterized by the F-H-T-R Ar/R filter. However, the four residues of the Ar/R filter were more diverse in other four subfamilies and showed some specificity within each subfamily or subgroup. Furthermore, the five Froger's positions were also indicated, which may work together with the Ar/R selectivity filter for the selectivity of some substrates (Hove and Bhave 2011). The residues at the P1 position were specific within each subfamily. Except for the SIP and XIP subfamilies, the members of other subfamilies feature the hydroxyl amino acid Thr or Ser at the P2 position. The P3 position features the Ala residue and the P4 position features the aromatic Try or Phe. All members have filled the P5 position with the Trp residue except for the NIP subfamily.

Besides, these general conserved sites, some important uncategorized residues were also indicated (Supplementary

S1). The closure of spinach *SoPIP2;1* was reported to be triggered by the dephosphorylation of two serine residues (Ser 115 and Ser 274) or protonation of a histidine (His 193) (Tornroth-Horsefield et al. 2006). The two serine residues were all conserved in PIP2 members, which was consistent with *Arabidopsis* as reported (Tornroth-Horsefield et al. 2006). Also, the histidine was conserved in all PIP sequences. Meanwhile, the conserved Leu 197 in *SoPIP2;1*, which was reported to be a key residue that combined with His 99, Val 104 and Leu 108 to create a hydrophobic barrier (Tornroth-Horsefield et al. 2006), was conserved in all PIP members. The position Ser 262 in soybean *GmNOD26* was reported to be related to the water permeability and regulated by osmotic signals (Guenther et al. 2003), the corresponding site of *MtAQPs* coincided with the Ser 274 site described above, and was conserved in most of the NIP members. Furthermore, the LGGC motif in loop C and NPARC motif in loop E were found in both XIP sequences, which were both deficient in other *AQPs* sequences (Supplementary S1) (Zou et al. 2015a, b). Moreover, these functional residues of *AQPs* in some aquaporins may also form a polymer to function, and the overexpression of one aquaporin may influence the expression of other aquaporins under different conditions (Jang et al. 2007).

In silico analysis of the expression patterns of *MtAQPs*

Six tissues and two common abiotic stresses (salt and drought) were selected for further expression analysis (Fig. 4 and Table S8). *MtNIP5;1* was excluded from the profiles due to the absence of the probe ID. For the expression of *MtAQPs* in different tissues (Fig. 4a), most of the highly expressed genes belong to PIP subfamily, especially *MtPIP2;1*, *MtPIP2;2* and *MtPIP2;3*, which can be found in all tissues with extremely high expression. Compared with other subfamily, the average expression of PIPs subfamily is approximately 4.19-fold higher. *MtTIP1;1* showed slightly lower expression in seed, but highly expressed in other tissues. These genes may play an important role in the constitutive transport functions. Besides, the higher expression of some genes in roots (*MtTIP1;2*, *MtTIP1;3*, *MtTIP1;4* and *MtTIP2;3*), stems (*MtTIP2;2*, *MtNIP6;1*, *MtNIP6;3* and *MtNIP6;4*), leaves (*MtNIP2;1*, *MtNIP4;1*, *MtNIP4;2*, *MtNIP4;3*, *MtSIP1;3* and *MtSIP2;1*), flowers (*MtNIP6;2*), seeds (*MtPIP2;4*, *MtPIP2;5*, *MtTIP3;1* and *MtTIP3;2*), nodules (*MtNIP1;5*), roots and flowers (*MtTIP2;1*) and seeds and nodules (*MtNIP1;6*) were identified, indicated their possible roles in corresponding tissues. Furthermore, *MtTIP3* showed the extremely high expression level in seeds when compared with other tissues, which was nearly 583-fold up-regulated. Earlier, a similar expression pattern for TIPs and more particularly TIP3s was reported in soybean and canola (Deshmukh et al. 2013; Sonah et al. 2017), this suggests a key role of TIP3s in regulating seed development. In *Arabidopsis* and pea, mercury derivatives reduced the rate of seed germination and seed imbibition, respectively, indicating aquaporins play a role in these processes (Veselova et al. 2003; Willigen et al. 2006). Negative correlation of TIP3s with other subfamilies is expected because of their tissue specificity. In our study, the expression of two SIPs subfamily genes *MtSIP1;1* and *MtSIP1;2* could be detected in all tissues, and approximately 3.98-fold up-regulated when compared with *MtSIP1;3* and *MtSIP2;1* genes.

Many stress responsive genes were reported to belong to PIP or TIP subfamily, however, few reports about specific functions of the members of NIP, SIP, and XIP subfamilies were found. In this study, we found that the most abiotic stresses responsive genes belong to NIP subfamily and have the drought correlative *cis*-element MBS (Fig. 4b; Table S6). Concretely, the gene *MtNIP1;6* may respond to both salt and drought in roots. Genes such as *MtPIP1;5*, *MtNIP6;2*, *MtNIP6;3* and *MtNIP6;4* may be induced by drought in both shoots and roots. However, there may be more uncertainty to the function of *MtNIP6;2*, because many functional residues of this gene are diverse. Besides, genes such as *MtTIP1;2*, *MtTIP1;3*, *MtTIP1;4*, *MtTIP2;2*, *MtNIP1;1*, *MtNIP1;2*, *MtNIP1;4*, *MtNIP1;5*, *MtNIP2;1* and *MtNIP7;1* may be salt responsive genes. Notably, *MtNIP7;1* likely worked as a salt responsive *AQPs* but another domain also was found, indicated the possibility of dual functions of this gene. Furthermore, genes such as *MtTIP3;1*, *MtNIP4;1*, *MtNIP4;2*, *MtNIP4;3* may have a role in response to drought stress in shoots. Meanwhile, except for *MtTIP3;1*, *MtNIP1;6* and *MtNIP6;3*, all these outstanding drought responsive genes have at least one drought related MBS element. For NIP subfamily members, clearly, the gene *TaNIP* of wheat was reported to enhance the salt tolerance in transgenic *Arabidopsis* (Gao et al. 2010). Besides, *AtNIP5;1*, *AtNIP6;1*, and *OsNIP3;1* were all reported to be crucial for the plant growth and development under boron limitation (Hanaoka et al. 2014; Takano et al. 2006; Tanaka et al. 2008). In this study, the NIP members showed some special structure features (Table 2). For instance, the residues at the P1 position were aromatic Tyr or Phe and the residues at the P5 position were different from the conserved Trp in other four subfamilies. In brief, NIP members may also play critical roles in response to some stresses and are worthy of further confirmation. Moreover, the XIP subfamily members may also have some functions unknown, such as the grapevine *VvXIP1*, which can be regulated by drought stress (Noronha et al. 2016).

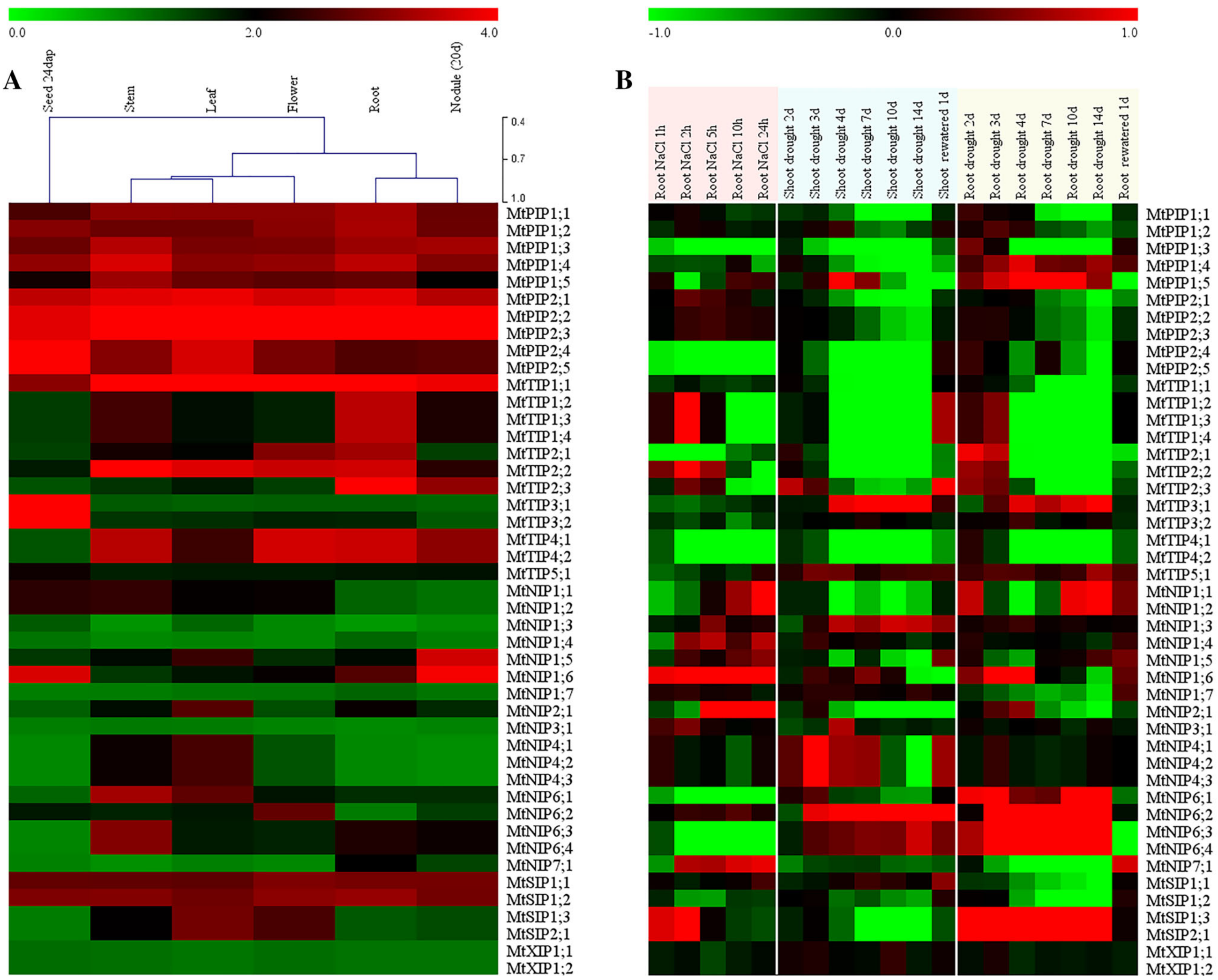

Fig. 4 Expression patterns of *MtAQPs* in different tissues and during different stress conditions. **a** The expression patterns of *MtAQPs* in seed, stem, leaf, flower, root and nodule tissues. **b** The expression patterns of each gene in root tissues of 2 weeks old seedlings under 200 mM NaCl stress, shoot, and root tissues under drought stress, respectively. Red color indicates up-regulation and green indicates down-regulation (colour figure online)

Validation of *MtAQP* gene expression patterns using qRT-PCR

Based on the silico analysis, ten *MtAQP* genes were selected and their expression profiles were analyzed under salt and drought treatments with qRT-PCR. Under salt conditions, *MtPIP1;5* and *MtTIP1;3* showed continuous up-regulation throughout the treatment. The expression of *MtNIP1;4* significantly induced and reached the highest expression level at 24 h, indicating its possible roles in salt stress responses. In contrast, the expression of *MtNIP1;6* and *MtNIP2;1* showed an opposite trend after salt treatment (Fig. 5). As shown in Fig. 5b, c, some genes showed similarly expression patterns between leaf and root under drought treatment. For example, *MtPIP1;5*, *MtTIP1;3* and *MtNIP7;1* were both significantly up-regulated compared with control in two tissues. The *MtNIP1;6* and *MtNIP6;4* showed opposite patterns in response to drought in leaves and roots. Both up and down regulation of *AQPs* in root and leaves have been reported previously and the expression with notable down-regulation in roots and up-regulation in leaves were highlighted (Galmés et al. 2007; Sonah et al. 2017). The contrasting response of *AQPs* in leaves and roots suggested that it may have different response mechanisms between these two tissues. Notably, *MtPIP1;5* and *MtTIP1;3* were significantly up-regulated under both drought and salinity stress conditions. These two outstanding drought and salinity responsive genes have three and five ARE elements, respectively, and one drought related MBS element. Two *AQPs* genes (*CaTIP1-4* and *CANIP4-1*), showed higher expression in both roots and shoot tissues under drought and salinity stresses contains

Fig. 5 The relative expression ratio of ten representative *MtAQP* genes in salinity and drought conditions. **a** Relative expression ratios salinity (under stress treatments for 0, 6, 24 and 48 h), **b**, **c** drought stress in leaf and root respectively (under stress treatments for 0, 6, 24 and 48 h) have been calculated with reference (*Actin* gene). Different capital letters indicate significant differences between different treatment time ($P < 0.05$). The name of the gene is written on the top of each bar diagram (error bars indicate the standard deviation from three replicates)

stress-responsive *cis*-acting elements MBS (Deokar and Tar'an 2016). The presence of MBS element may be play important roles in these two stresses. Overexpression of *BnPIP1* in transgenic tobacco plants resulted in an increased tolerance to water stress at the whole plant level (Gao et al. 1999). Transgenic *Arabidopsis* lines over-expressing *JcPIP2;7* and *JcTIP1;3* shown improved germination and seed viability under high salt and mannitol (Khan et al. 2015). The overexpression of *TaNIP* up-regulated several stress-associated genes and enhanced the salt tolerance in transgenic *Arabidopsis*, indicate that it plays an important role in salt tolerance in *Arabidopsis* and can also enhance plants' tolerance to other abiotic stresses (Gao et al. 2010). Therefore, these two genes could be selected as excellent candidates for further functional characterization and application in legumes breeding programs.

In conclusion, a total of 46 structurally complete *MtAQPs* were identified, results showed that two proteins (MtNIP6;2 and MtNIP7;1) possessed the special domains, compared to the common *AQPs* family with the other species and showed some of difference natures, and those two proteins may exhibit the corresponding functions but needed further confirmation. The *cis*-elements were found in promoter sequences of *MtAQPs*, which provide potential possibilities for the functions of these genes. Combined with the expression analysis and qRT-PCR testing, *MtPIP1;5* and *MtTIP1;3* were significantly up-regulated under both drought and salinity stress conditions, which indicated that these members may be the main operative proteins in general conditions and play vital roles response to abiotic stresses.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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